

Exercise 4

Task 1 - Energy Balance

(a) Energy Balance over a Snow Surface

The energy balance of a snowpack describes the different energy fluxes entering and leaving the snow surface. The energy-balance model used in this exercise expresses the balance as:

$$Q_m + Q_i = Q_s + Q_l + Q_h + Q_e + Q_g + Q_r$$

Where Q_m is the energy available for snowmelt, Q_i is the energy required to change the temperature of the snowpack, Q_s is net short-wave radiation, Q_l is net long-wave radiation, Q_h is sensible heat transfer, Q_e is latent heat transfer, Q_g is heat exchange with the ground, and Q_r is the heat content of precipitation.

For a simplified snowmelt calculation, the energy available for melting can therefore be expressed as the combined contribution of the main surface energy fluxes:

$$Q_m = Q_s + Q_l + Q_h + Q_e + Q_g + Q_r - Q_i$$

The net radiation component consists of short-wave and long-wave radiation:

$$R_n = Q_s + Q_l$$

Where Q_s represents the balance between incoming and reflected solar radiation, while Q_l represents the balance between incoming atmospheric and outgoing surface long-wave radiation.

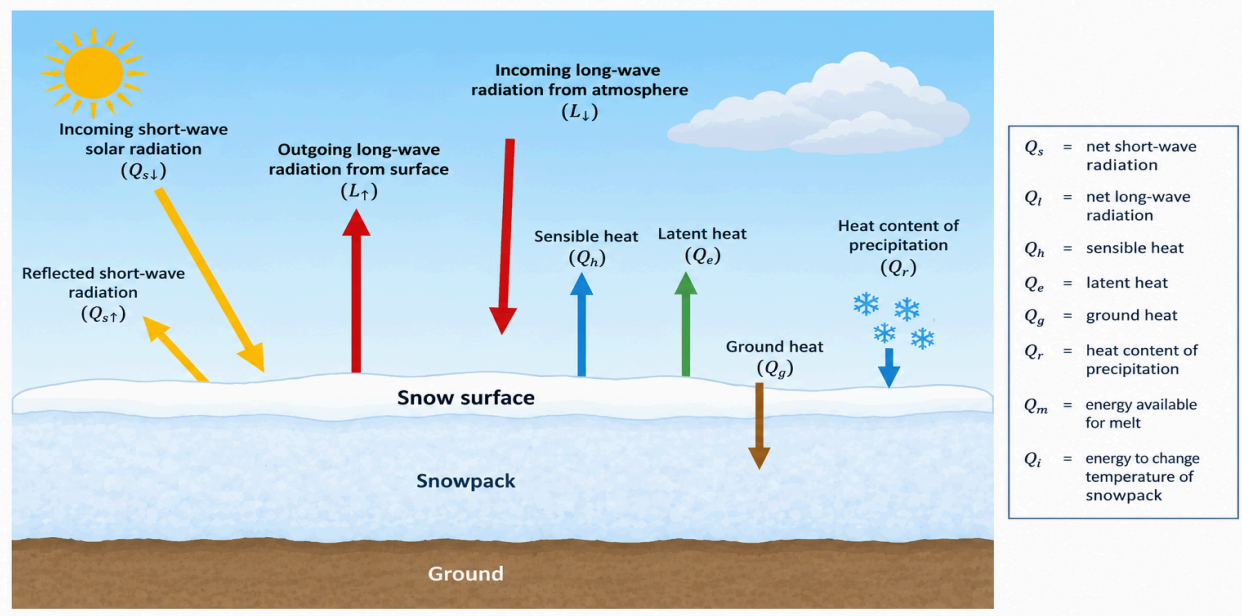


Figure 1. Schematic representation of the main energy fluxes involved in the energy balance of a snow-covered surface.

(b) Long-Wave Radiation from the Snow Surface

The surface temperature is:

$$T_0 = 10^\circ\text{C}$$

Converting to Kelvin:

$$T_0 = 10 + 273.15 = 283.15 \text{ K}$$

The outgoing long-wave radiation is calculated using the Stefan–Boltzmann equation:

$$L_{\uparrow} = \varepsilon \sigma T_0^4$$

where ε is the surface emissivity and

$$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$$

For an approximately blackbody surface, $\varepsilon \approx 1$. Therefore:

$$L\uparrow = 5.67 \times 10^{-8} (283.15)^4$$

$$L\uparrow \approx 373.5 \text{ W m}^{-2}$$

Thus, the surface emits approximately 374 W m⁻² of long-wave radiation.

This outgoing radiation becomes a net long-wave loss when the emitted radiation exceeds the incoming atmospheric long-wave radiation:

$$L\uparrow > L\downarrow$$

or:

$$L_{net} = L\downarrow - L\uparrow < 0$$

The magnitude of this net long-wave loss depends mainly on atmospheric temperature, cloud cover and atmospheric water-vapour content. Clouds and water vapour increase atmospheric long-wave back radiation towards the snow surface and therefore reduce the net long-wave loss. Clear, cold and dry conditions generally result in a larger net long-wave loss.

The incoming atmospheric long-wave radiation can be represented conceptually as:

$$L\downarrow \approx \epsilon_a \sigma T_a^4$$

where ϵ_a is the effective atmospheric emissivity and T_a is the effective atmospheric temperature. This illustrates the role of the greenhouse effect: atmospheric gases and clouds absorb terrestrial long-wave radiation and re-emit part of it towards the surface.

Task 2 - Snow Water Equivalent

The snow sample was collected using a tube with a diameter of 10 cm. The empty tube had a mass of 1.1 kg, while the total mass of the tube and snow was 2.600 kg. The measured snow depth was 55 cm.

The mass of snow is therefore:

$$m_s = 2.600 - 1.100 = 1.500 \text{ kg}$$

The radius of the tube is:

$$r = \frac{0.10}{2} = 0.05 \text{ m}$$

and its cross-sectional area is:

$$A = \pi r^2$$

$$A = \pi(0.05)^2 = 0.007854 \text{ m}^2$$

The equivalent volume of liquid water is:

$$V_w = \frac{m_s}{\rho_w}$$

where $\rho_w = 1000 \text{ kg m}^{-3}$. Thus:

$$V_w = \frac{1.500}{1000} = 0.001500 \text{ m}^3$$

The water-equivalent depth is:

$$h_w = \frac{V_w}{A}$$

$$h_w = \frac{0.001500}{0.007854} = 0.1910 \text{ m}$$

Therefore:

$$SWE \approx 191 \text{ mm}$$

As a consistency check, the snow volume is:

$$V_s = Ah_s$$

$$V_s = 0.007854 \times 0.55 = 0.004320 \text{ m}^3$$

giving a mean snow density of:

$$\rho_s = \frac{1.500}{0.004320}$$

$$\rho_s \approx 347 \text{ kg m}^{-3}$$

The calculated snow density gives the same water equivalent of approximately 191 mm.

Task 3 - Snow Melt

(a) Energy-Balance Model

Daily snowmelt was calculated using the supplied **SMELT_99 energy-balance model**. The model was configured according to the exercise instructions: the first day was set to 1 April, the latitude was set to 63.5° N, the snow-cover age was set to 30 days on the first day, and the wind-gauge height was set to 2 m.

The meteorological input data were:

Day	Air temperature (°C)	Relative humidity (%)	Wind speed (m/s)	Cloud cover
1	3.6	80	1.1	0.6
2	2.2	78	0.3	0.6
3	1.0	76	0.4	0.1
4	2.6	80	0.4	0.6
5	3.7	93	0.4	0.8
6	3.0	96	0.4	0.8
7	3.6	85	0.4	0.9

The resulting daily snowmelt calculated by the energy-balance model was:

Day	Energy-balance melt (mm/day)
1	9.61
2	3.76
3	0.00
4	5.61
5	10.62
6	9.15
7	9.33

The total simulated melt over the seven-day period was:

48.08 mm

The model therefore predicts substantial melt on most days, while no melt occurs on Day 3 because the combined energy balance is insufficient to produce positive melt.

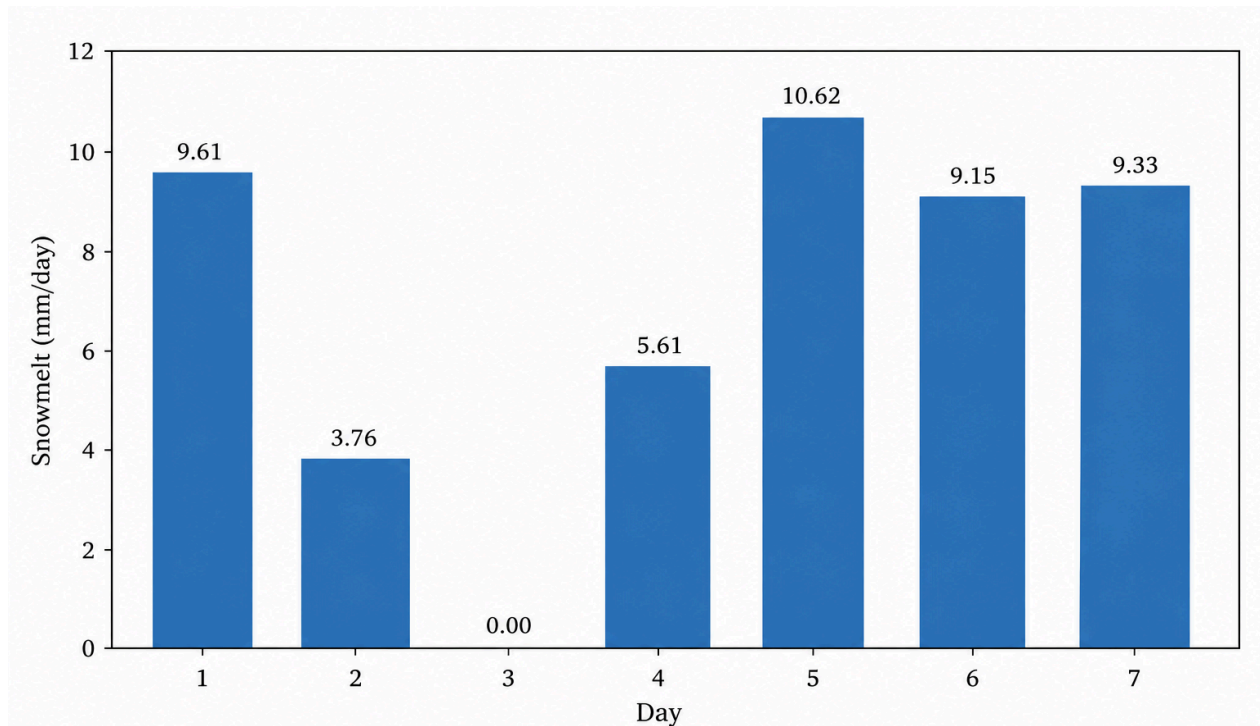


Figure 2. Daily snowmelt calculated using the energy-balance model for the seven-day period beginning on 1 April.

(b) Degree-Day Model

A degree-day or temperature-index model was subsequently used to reproduce the snowmelt calculated by the energy-balance model. The model is expressed as:

$$M = C(T - ts)$$

when:

$$T > ts$$

and:

$$M = 0$$

when:

$$T \leq t_s$$

where M is daily snowmelt, C is the degree-day factor, T is air temperature and t_s is the threshold temperature.

The values of C and t_s were kept constant for the entire seven-day period, as required by the exercise.

The best fit to the energy-balance results was obtained with:

$$C = 4.116 \text{ mm} \cdot \text{day}^{-1} \cdot ^\circ\text{C}^{-1}$$

and:

$$t_s = 1.17^\circ\text{C}$$

Using these values gives:

Day	Air temperature (°C)	Energy-balance melt (mm/day)	Degree-day melt (mm/day)
1	3.6	9.61	10.00
2	2.2	3.76	4.24
3	1.0	0.00	0.00
4	2.6	5.61	5.89
5	3.7	10.62	10.41
6	3.0	9.15	7.53
7	3.6	9.33	10.00

The degree-day model reproduces the general temporal pattern of the energy-balance model, although differences remain on individual days. The largest discrepancy occurs on Day 6, when the degree-day model gives 7.53 mm/day compared with 9.15 mm/day from the energy-balance model.

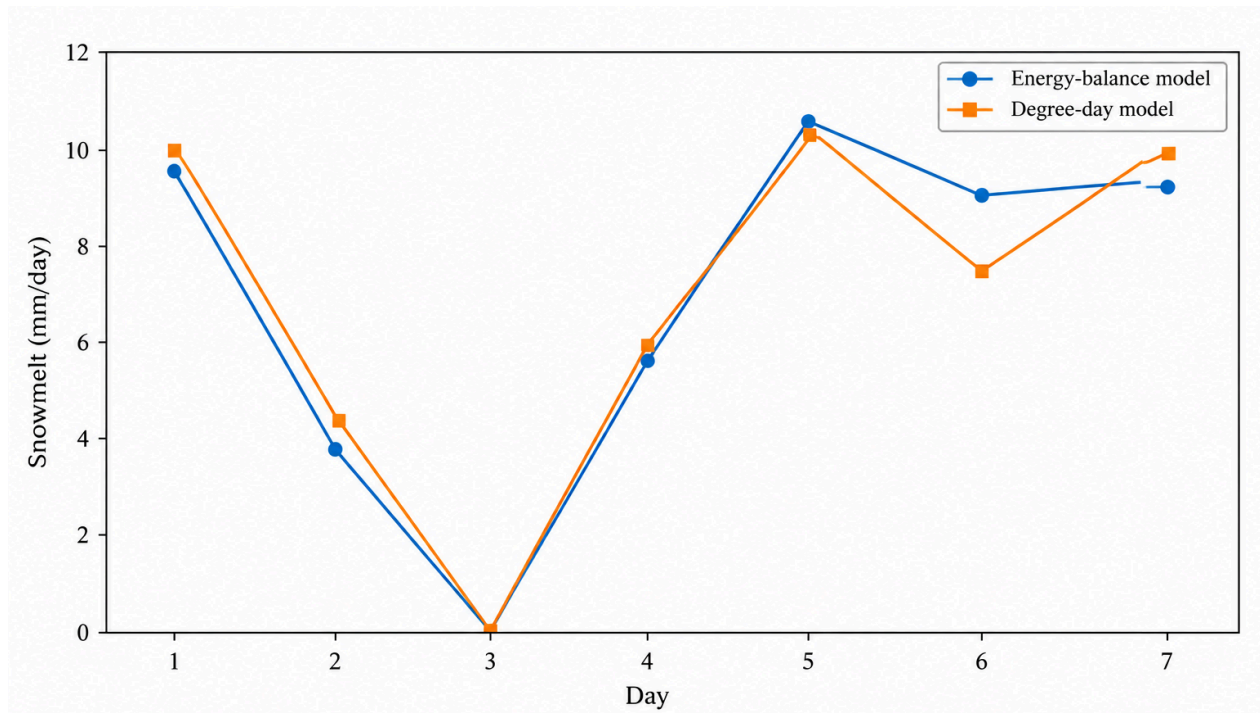


Figure 3. Comparison between daily snowmelt calculated using the energy-balance model and the calibrated degree-day model. The degree-day model uses a constant degree-day factor of 4.116 mm day⁻¹ °C⁻¹ and a threshold temperature of 1.17 °C.

(c) Evaluation of the Calibration

The agreement between the energy-balance and degree-day estimates was evaluated using the Pearson correlation coefficient, as suggested in the exercise.

The correlation coefficient was:

$$r = 0.980$$

The corresponding coefficient of determination is:

$$R^2 = 0.961$$

The root-mean-square error was:

$$RMSE = 0.714 \text{ mm/day}$$

The high correlation indicates that the degree-day model reproduces the temporal variation in the energy-balance snowmelt very well over the analysed period. The relatively low RMSE further indicates that the differences between the two approaches are generally modest.

However, the degree-day model does not reproduce every daily variation exactly. This is expected because the energy-balance model explicitly accounts for several meteorological controls, including radiation, humidity and wind, whereas the degree-day model represents the melt response primarily through air temperature.

The results therefore show that a simple temperature-index model can provide a good approximation of the energy-balance melt over this short period when appropriate constant values of C and t_s are selected.

Task 4 - Conclusion

The energy balance of a snow surface is controlled by several interacting energy fluxes, including short-wave and long-wave radiation, sensible and latent heat transfer, ground heat exchange and the heat content of precipitation. For a surface temperature of $10\text{ }^{\circ}\text{C}$, the outgoing long-wave radiation is approximately 373.5 W m^{-2} . The net long-wave loss depends strongly on atmospheric conditions, particularly temperature, humidity and cloud cover.

The snow-sampling calculation produced a snow water equivalent of approximately 191 mm, corresponding to a mean snow density of approximately 347 kg m^{-3} .

For the seven-day period beginning on 1 April, the SMELT_99 energy-balance model produced a total simulated snowmelt of approximately 48.08 mm. A degree-day model calibrated against these energy-balance results gave a degree-day factor of $4.116\text{ mm day}^{-1}\text{ }^{\circ}\text{C}^{-1}$ and a threshold temperature of $1.17\text{ }^{\circ}\text{C}$.

The calibrated degree-day model achieved a correlation coefficient of 0.980, an R^2 of 0.961, and an RMSE of 0.714 mm/day . This indicates a strong correspondence between the two modelling approaches over the analysed period. Nevertheless, some daily differences remain because the degree-day method does not explicitly represent the individual effects of radiation, humidity and wind.